

D831

Introduction to AI and Security

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

At a trade show with 300 exhibitors, a vendor says its security tool uses deep learning. Which statement about deep learning is accurate?

- A. It is separate from machine learning.
- B. All AI is deep learning.
- C. AI is a subset of deep learning.
- D. It is a subset of machine learning, which is a subset of AI.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A spam filter runs only on hand-written if-then rules from analysts and never learns from data. How is it classified?

- A. Deep learning
- B. Rule-based AI, but not machine learning
- C. Machine learning, because it filters automatically
- D. Not AI at all, since AI always learns from data

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An intrusion detection model was trained on network traffic from a 500-person office in 2023. After the company moves to cloud services and remote work, the model's accuracy drops sharply. What is the most likely cause?

- A. The model must have been tampered with.
- B. Distribution shift; the model needs monitoring and retraining on current traffic.
- C. Validated models never lose accuracy, so the data must be wrong.
- D. The office was too small for AI to work.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An analyst at a 75-person firm asks a generative AI chatbot with a polished interface to summarize a new vendor advisory. The answer confidently cites a specific vulnerability ID and patch version. What should the analyst do?

- A. Act on it, because specific details show it is accurate.
- B. Trust it, since the interface is professional.
- C. Check the ID and patch version against the vendor's official advisory before acting.
- D. Stop using AI tools entirely.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which description best defines a generative AI hallucination?

- A. Output an outsider deliberately planted through the training data
- B. A system crash during generation
- C. Fluent, plausible output that is false or unsupported by any source
- D. Errors that occur only when prompts contain typos

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A resume-screening model was trained on ten years of a company's past hiring decisions, when most hires came from three universities. The company has offices in five states. The model now ranks graduates of other schools lower. Why?

- A. Having offices in five states confused the model.
- B. It learned the patterns in past hiring, including its bias.
- C. Algorithms are objective, so the ranking reflects merit.
- D. Removing the university field would guarantee fairness.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A company connects an AI assistant to employee email with permission to send messages and delete files. An incoming email hides text telling the assistant to forward files to an outside address. The company's email provider advertises 99.9% uptime. What is the best defensive design?

- A. Switch to an email provider with higher uptime.
- B. Rely on the model to ignore instructions it finds in emails.
- C. Give the assistant more permissions so it can quarantine bad emails itself.
- D. Treat email content as untrusted data, limit the assistant to least privilege, and require human approval to send or delete.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which design principle most limits the damage if an AI agent is manipulated by malicious input?

- A. Least privilege, with human approval for high-impact actions
- B. A larger, more capable model
- C. A keyword filter, since filters always stop manipulation
- D. Full admin rights so the agent works efficiently

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A malware classifier retrains every night on file samples that anyone can submit through a public web form. The team of 12 analysts stores samples in two cloud regions. A security review flags a risk. What is it, and what is the fix?

- A. Data poisoning; vet and track the source of samples before training and watch for unusual shifts.
- B. Storing in two regions; consolidate to one.
- C. Retrain more often to dilute bad samples.
- D. Evasion at prediction time; add more features.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A security team at a retailer with 1,200 stores runs an anomaly detector on 1,000,000 events a day. About 50 events a day are truly malicious. The detector flags 98% of malicious events and 1% of benign ones. About how many false alarms does it raise per day?

- A. About 10,000
- B. 49
- C. About 10,049
- D. 0.5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A detector raised 10,049 alerts in a day, and 49 were real incidents. What share of its alerts were real?

- A. 99%
- B. 100%, since the detector flags only anomalies
- C. 98%
- D. About 0.5%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A bank with 3,000 employees launched an AI model that sets credit limits after a one-time review last year. Its headquarters is LEED-certified. The board asks how the bank should govern AI risk. What is the best approach?

- A. Leave governance to the data scientists who built it.
- B. Rely on the building's LEED certification for sustainability oversight.
- C. Assign accountability, keep a model inventory, and measure performance and bias on an ongoing basis with human oversight.
- D. The pre-launch review is sufficient.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

~~At a trade show with 300 exhibitors,~~ a vendor says its security tool uses deep learning. Which statement about deep learning is accurate?

Target word: how the terms nest **Decides:** DL $\subset$ ML $\subset$ AI.

A	Confusion set	Deep learning is one kind of machine learning.
B	Absolute language	Rule-based systems and classic ML are AI too.
C	Inverted operation	Reverses the nesting.
D	Correct	Deep learning is machine learning with many-layered neural networks.

2. Answer B Concept 1 · transfer

A spam filter runs only on hand-written if-then rules from analysts and never learns from data. How is it classified?

Target word: reverse: from the system to the category **Decides:** Rules, no learning.

A	Adjacent category	No neural network is involved.
B	Correct	It automates judgment without learning from data.
C	Confusion set	Automation is not learning.
D	Absolute language	Expert systems are a classic form of AI.

3. Answer B Concept 2

An intrusion detection model was trained on network traffic from a ~~500-person~~ office in 2023. After the company moves to cloud services and remote work, the model's accuracy drops sharply. What is the most likely cause?

Target word: why accuracy dropped **Decides:** The traffic now differs from the training data.

A	Adjacent category	Possible, but the environment change explains the drop directly.
B	Correct	Models only hold up on data like what they learned from.
C	Absolute language	Models drift as conditions change.
D	Camouflage	Office size does not decide the drift.

4. Answer C Concept 3

An analyst at a ~~75-person~~ firm asks a generative AI chatbot ~~with a polished interface~~ to summarize a new vendor advisory. The answer confidently cites a specific vulnerability ID and patch version. What should the analyst do?

Target word: how to use the output **Decides:** Generative output can be confidently wrong.

A	Confusion set	Specificity is not accuracy; hallucinations are often specific.
B	Camouflage	Design polish does not decide accuracy.
C	Correct	Verification catches hallucinated details.
D	Absolute language	AI can help when outputs are verified.

5. Answer C Concept 3 · transfer

Which description best defines a generative AI hallucination?

Target word: reverse: from term to definition **Decides:** Plausible but false output.

A	Confusion set	That is poisoning, not hallucination.
B	Adjacent category	A failure to run, not false content.
C	Correct	That is the defining feature.
D	Absolute language	Hallucinations occur with clean prompts too.

6. Answer B Concept 4

A resume-screening model was trained on ten years of a company's past hiring decisions, when most hires came from three universities. ~~The company has offices in five states.~~ The model now ranks graduates of other schools lower. Why?

Target word: source of the skew **Decides:** Trained on past human decisions.

A	Camouflage	Office count does not decide the bias.
B	Correct	Models reproduce the data they are trained on.
C	Absolute language	Models inherit bias from data.
D	Dropped element	Other fields can act as proxies.

7. Answer D Concept 5

A company connects an AI assistant to employee email with permission to send messages and delete files. An incoming email hides text telling the assistant to forward files to an outside address. ~~The company's email provider advertises 99.9% uptime.~~ What is the best defensive design?

Target word: defensive control for AI agents **Decides:** The assistant acts on untrusted content with broad permissions.

A	Camouflage	Uptime does not decide the risk.
B	Dropped element	Drops any control outside the model.
C	Inverted operation	Widens the damage if it is manipulated.
D	Correct	Limits what manipulated input can cause.

8. Answer A Concept 5 · transfer

Which design principle most limits the damage if an AI agent is manipulated by malicious input?

Target word: reverse: from goal to principle **Decides:** Damage is bounded by what the agent may do.

A	Correct	Bounds what any manipulated action can reach.
B	Adjacent category	Capability does not bound permissions.
C	Absolute language	Filters help but can be bypassed.
D	Inverted operation	Maximizes potential damage.

9. Answer A Concept 6

A malware classifier retrains every night on file samples that anyone can submit through a public web form. ~~The team of 12 analysts stores samples in two cloud regions.~~ A security review flags a risk. What is it, and what is the fix?

Target word: risk and control **Decides:** Untrusted data flows straight into training.

A	Correct	Protecting training data integrity blocks poisoning.
B	Camouflage	Storage regions do not decide the risk.
C	Inverted operation	Faster retraining takes in bad data faster.
D	Confusion set	Evasion targets a trained model; the flaw here is in training data.

10. Answer A Concept 7

A security team ~~at a retailer with 1,200 stores~~ runs an anomaly detector on 1,000,000 events a day. About 50 events a day are truly malicious. The detector flags 98% of malicious events and 1% of benign ones. About how many false alarms does it raise per day?

Target word: false positives per day **Decides:** 1% of 999,950 benign events.

A	Correct	$999,950 \times 0.01 \approx 10,000$.
B	Confusion set	The true detections (50×0.98).
C	Adjacent category	Total alerts, including the real ones.
D	Inverted operation	Applies the 1% rate to malicious events instead of benign ones.

11. Answer D Concept 7 · transfer

A detector raised 10,049 alerts in a day, and 49 were real incidents. What share of its alerts were real?

Target word: solve for alert precision **Decides:** $49 \div 10,049$.

A	Inverted operation	1 minus the false-positive rate; not alert precision.
B	Absolute language	Most anomalies are benign.
C	Confusion set	The detection rate, not the share of alerts that are real.
D	Correct	$49 \div 10,049 \approx 0.0049$.

12. Answer C Concept 8

A bank ~~with 3,000 employees~~ launched an AI model that sets credit limits after a one-time review last year. ~~Its headquarters is LEED-certified.~~ The board asks how the bank should govern AI risk. What is the best approach?

Target word: sound AI governance **Decides:** A one-time review of a model now in production.

A	Adjacent category	Governance needs independent accountability and business owners.
B	Camouflage	A building rating does not govern models.
C	Correct	Governance runs across the whole lifecycle.
D	Dropped element	Drops ongoing monitoring; models drift.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	8	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	AI, machine learning, and deep learning	Deep learning is a subset of machine learning, which is a subset of AI; rule-based systems are AI without learning from data.	Treating the terms as separate or reversing the nesting.	A vendor claim using one of the three terms.
2	Narrow AI and distribution shift	A model performs well only on data like its training data; when the environment changes, accuracy drifts and the model must be monitored and retrained.	Assuming a validated model stays accurate forever.	Performance drops after a change in the environment.
3	Generative AI hallucination	Generative models can produce fluent, specific, confident output that is false, so outputs must be verified against authoritative sources.	Reading confidence or detail as accuracy.	An AI answer with precise numbers or citations used for a decision.
4	Bias from training data	Models learn patterns in historical data, including past bias; removing one attribute does not remove proxies.	Believing algorithms are objective by nature.	A model trained on past human decisions.
5	Securing AI systems and agents	Treat external content as untrusted data, give AI tools least privilege, and require human approval for high-impact actions.	Trusting the model to ignore embedded instructions or giving it broad permissions.	An AI assistant with access to email, files, or actions.
6	Training data integrity	Protect the provenance of training data; unvetted data sources let bad data poison a model.	Confusing poisoning (training time) with evasion (prediction time).	A model that retrains on data from untrusted sources.
7	AI in security operations: alert math	A small false-positive rate on huge event volumes produces far more false alarms than real ones, so AI triage needs tuning and human review.	Reading the detection rate as the share of alerts that are real.	Event volume, a rare malicious rate, and a false-positive rate.
8	AI governance across the lifecycle	Govern AI continuously: assign accountability, inventory models, measure performance and bias, and keep human oversight (as in the NIST AI RMF's govern, map, measure, manage).	Treating a one-time pre-launch review as governance.	A model in production with no ongoing review.

Course scope (public description): Introduction to AI and Security provides an overview of critical terminology and key concepts for artificial intelligence (AI). Students explore the history, development, and evolution of AI along with its limitations and ethical issues. Students further explore best practices for securing AI within professional environments. Contextualized examples offer students an opportunity explore how AI is used in security operations.